Problem 1

$$J_a(x) = \frac{1}{2}\int_0^{t_f} [x^2(t) + u^2(t)]dt + \int_0^{t_f} [\lambda(f(u,x) - x'(t))]dt$$

where

$$f = -x + u$$

$$\lambda(t) = \lambda_0(t)$$

$$L = \frac{1}{2}(x^2(t) + u^2(t))$$

Then:

$$J_a(x) = \int_0^{t_f} [\lambda(f(u,x) - x'(t))]dt$$

where

$$H = L + \lambda f(u,x)$$

Then:

$$J_a(x) = \int_0^{t_f} [H - \lambda x'(t)]dt$$

First order optimality condition:

$$\delta J_a(x) = \delta \int_0^{t_f} [H - \lambda x'(t)]dt = 0$$

$$\delta J_a(x) = \int_0^{t_f} [\delta H - \delta\lambda x'(t) - \lambda\delta x'(t)]dt = 0$$

$$\delta J_a(x) = \int_0^{t_f} [\frac{\partial H}{\partial u}\delta u + \frac{\partial H}{\partial x}\delta x + \frac{\partial H}{\partial \lambda}\delta\lambda - \delta\lambda x'(t) - \lambda\delta x'(t)]dt = 0$$

$$\delta H = \frac{\partial H}{\partial u}\delta u + \left(\frac{\partial H}{\partial \underline{x}} + \dot{\underline{\lambda}}\right)\delta \underline{x} + \left(\frac{\partial H}{\partial \underline{\lambda}} - \dot{\underline{x}}\right)\delta \underline{\lambda} = 0$$

Since δu, $\delta \underline{x}$, $\delta \underline{\lambda}$ are independent, each term must vanish separately:

$$\frac{\partial H}{\partial u} = 0$$

$$\frac{\partial H}{\partial \underline{x}} + \dot{\underline{\lambda}} = 0 \implies \dot{\underline{\lambda}} = -\frac{\partial H}{\partial \underline{x}}$$

$$\frac{\partial H}{\partial \underline{\lambda}} - \dot{\underline{x}} = 0 \implies \dot{\underline{x}} = \frac{\partial H}{\partial \underline{\lambda}}$$

Given:

$$H = \frac{1}{2}(x^2 + u^2) + \lambda(-x + u)$$

Stationarity condition:

$$\frac{\partial H}{\partial u} = u + \lambda = 0 \implies u = -\lambda$$

Costate equation:

$$\dot{\lambda} = -\frac{\partial H}{\partial x} = -\frac{\partial}{\partial x}\left[\frac{1}{2}x^2 + \lambda(-x + u)\right] = -(x - \lambda) = -x + \lambda$$

$$\implies \dot{\lambda} = -x + \lambda$$

State equation:

$$\dot{x} = \frac{\partial H}{\partial \lambda} = -x + u$$

Substituting $u = -\lambda$:

$$\dot{x} = -x - \lambda$$

Writing as a matrix system:

$$\begin{pmatrix} \dot{x} \\ \dot{\lambda} \end{pmatrix} = \begin{pmatrix} -1 & -1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} x \\ \lambda \end{pmatrix}$$

At this point we also apply our given Boundary Conditions:

$$x(0) = x_0 = 1$$

$$x(t_f) = x_f = 1$$

$$t_f = fixed$$

Thus, we must guess λ_0 and then perform integration from t_0 to t_f for simple shooting. The analytical solution is included below the simple shooting implementation

Simple shooting implementation

In [46]:
```
#General python includes that we will be using
import numpy as np
from scipy.integrate import solve_ivp
from scipy.optimize import fsolve
import warnings
#I want to not display warnings for stuff that doesn't matter
warnings.filterwarnings('ignore', message='Creating an ndarray from ragged nested s
```

In [47]:
```
A = np.array([[-1, -1],
              [-1,  1]])

def system(t, y):
    return A @ np.array(y)



def integrator(guess, end_time):
    #solve_ivp input:
    #0: func handle with time as first parameter
    #1: time start : time end
    #2: initial value_1 : initial value_2
    # flag: dense_output if we want all t's not just endpoint
    sol = solve_ivp(system, np.array([0, end_time]), np.array([1, guess]),
                    method='DOP853',
                    rtol=1e-8,
                    atol=1e-8,
                    dense_output=True)
    t = np.linspace(0, end_time, int(10000*end_time ))
    y = sol.sol(t)

    x = y[0]
    lam = y[1]
    return x, lam, t
```

In [48]:
```
def error(guess, time):
    x, lam, t = integrator(guess, time)
    return x[-1] - 1

import matplotlib.pyplot as plt

def solve_loop(time_array):
    guess = 0
    for time in time_array:
        solution, info, ier, msg = fsolve(error, guess, args=(time,), xtol=1e-8, fu
        lambda_0 = solution[0]
```

```
        x, lam, t = integrator(lambda_0, time)

        # u(t) = -lambda(t)
        u = -lam

        print(f"tf={time}: lambda(0) = {lambda_0:.10f}, x(tf) = {x[-1]:.10f}")
        print(f"u(0) = {u[0]:.10f}, u(tf) = {u[-1]:.10f}")
        print(f"Converged: {ier==1}, message: {msg}\n")

        guess = lambda_0
        #guess = 0
```

In [49]:
```
time_array = [10, 20, 30, 40, 50]
solve_loop(time_array)
```

```
tf=10: lambda(0) = 0.4142115221, x(tf) = 1.0000000002
u(0) = -0.4142115221, u(tf) = 2.4142115225
Converged: True, message: The solution converged.

tf=20: lambda(0) = 0.4142135624, x(tf) = 1.0002733102
u(0) = -0.4142135624, u(tf) = 2.4148733916
Converged: True, message: The solution converged.

tf=30: lambda(0) = 0.4142135624, x(tf) = 181.4539984517
u(0) = -0.4142135624, u(tf) = 438.0687040089
Converged: True, message: The solution converged.

tf=40: lambda(0) = 0.4142135624, x(tf) = 251546337.4230465889
u(0) = -0.4142135624, u(tf) = 607286579.3719977140
Converged: True, message: The solution converged.

tf=50: lambda(0) = 0.4142135624, x(tf) = 348714056780004.8750000000
u(0) = -0.4142135624, u(tf) = 841870205268429.6250000000
Converged: True, message: The solution converged.
```

Analytical solution derivation

Eigenvalues

Solve: $\det(A - \sigma I) = 0$

$$(-1-\sigma)(1-\sigma) - 1 = 0$$

$$\sigma^2 - 2 = 0$$

Thus: $\sigma = \pm\sqrt{2}$

Eigenvectors

For $\sigma = \sqrt{2}$:

$$v_1 = \begin{bmatrix} 1 \\ -(1+\sqrt{2}) \end{bmatrix}$$

For $\sigma = -\sqrt{2}$:

$$v_2 = \begin{bmatrix} 1 \\ \sqrt{2}-1 \end{bmatrix}$$

General Solution

$$x(t) = c_1 e^{\sqrt{2}\,t} + c_2 e^{-\sqrt{2}\,t}$$

$$\lambda(t) = -(1+\sqrt{2})\,c_1 e^{\sqrt{2}\,t} + (\sqrt{2}-1)\,c_2 e^{-\sqrt{2}\,t}$$

$$u(t) = -\lambda(t)$$

through boundary conditions:

$$x(0) = 1, \qquad x(t_f) = 1$$

So:

$$c_1 + c_2 = 1$$

$$c_1 e^{\sqrt{2}\,t_f} + c_2 e^{-\sqrt{2}\,t_f} = 1$$

Solve constants after choosing each t_f.

Solving for c_1, c_2, and $u(t)$ for Each t_f

System to solve (general):

$$c_1 + c_2 = 1$$

$$c_1 e^{\sqrt{2}\,t_f} + c_2 e^{-\sqrt{2}\,t_f} = 1$$

From the first equation: $c_2 = 1 - c_1$. Substituting:

$$c_1 e^{\sqrt{2}\,t_f} + (1-c_1) e^{-\sqrt{2}\,t_f} = 1$$

$$c_1 \left(e^{\sqrt{2}\,t_f} - e^{-\sqrt{2}\,t_f} \right) = 1 - e^{-\sqrt{2}\,t_f}$$

$$\boxed{c_1 = \frac{1 - e^{-\sqrt{2}\,t_f}}{e^{\sqrt{2}\,t_f} - e^{-\sqrt{2}\,t_f}}, \qquad c_2 = 1 - c_1}$$

Then:

$$u(t) = -\lambda(t) = (1+\sqrt{2})\,c_1 e^{\sqrt{2}\,t} - (\sqrt{2}-1)\,c_2 e^{-\sqrt{2}\,t}$$

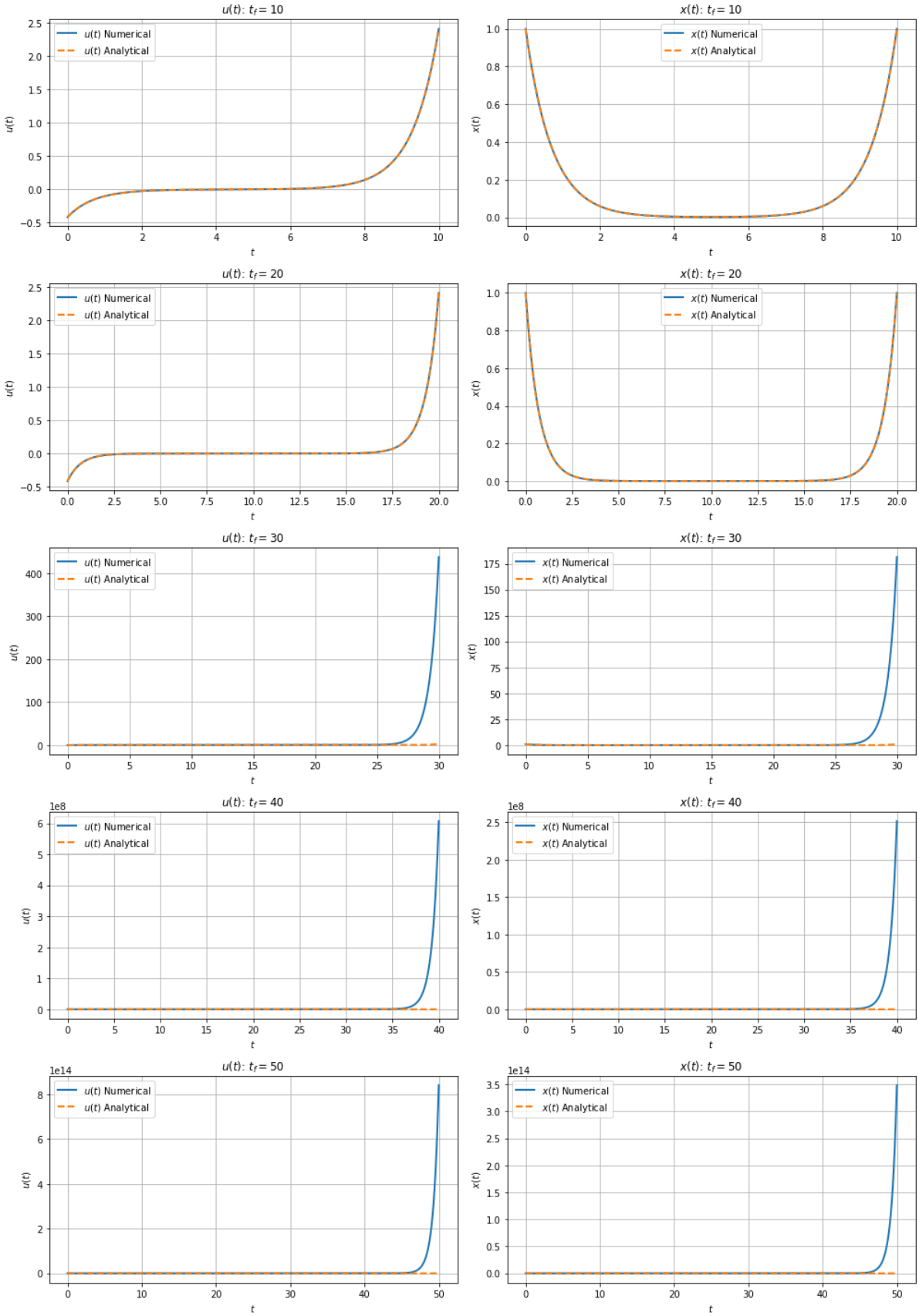
$$x(t) = c_1 e^{\sqrt{2}\,t} + c_2 e^{-\sqrt{2}\,t}$$

Comparison

In [50]:
```
def analytical_u(t, tf):
    sqrt2 = np.sqrt(2)
    c1 = (1 - np.exp(-sqrt2 * tf)) / (np.exp(sqrt2 * tf) - np.exp(-sqrt2 * tf))
    c2 = 1 - c1
    return (1 + sqrt2) * c1 * np.exp(sqrt2 * t) - (sqrt2 - 1) * c2 * np.exp(-sqrt2

def analytical_x(t, tf):
    sqrt2 = np.sqrt(2)
    c1 = (1 - np.exp(-sqrt2 * tf)) / (np.exp(sqrt2 * tf) - np.exp(-sqrt2 * tf))
    c2 = 1 - c1
    return c1 * np.exp(sqrt2 * t) + c2 * np.exp(-sqrt2 * t)

def solve_and_compare(time_array):
    guess = 0
    fig, axes = plt.subplots(len(time_array), 2, figsize=(14, 4*len(time_array)))

    for i, tf in enumerate(time_array):
        solution, info, ier, msg = fsolve(error, guess, args=(tf,), xtol=1e-8, full
        lambda_0 = solution[0]
        x, lam, t = integrator(lambda_0, tf)

        u_numerical  = -lam
        u_analytical = analytical_u(t, tf)
        x_analytical = analytical_x(t, tf)

        # u(t) plots
        axes[i, 0].plot(t, u_numerical,  label='$u(t)$ Numerical',  linewidth=2)
        axes[i, 0].plot(t, u_analytical, label='$u(t)$ Analytical', linewidth=2, li
        axes[i, 0].set_title(f'$u(t)$: $t_f = {tf}$')
        axes[i, 0].set_xlabel('$t$')
        axes[i, 0].set_ylabel('$u(t)$')
        axes[i, 0].legend()
        axes[i, 0].grid(True)

        # x(t) plots
        axes[i, 1].plot(t, x,          label='$x(t)$ Numerical',  linewidth=2)
        axes[i, 1].plot(t, x_analytical, label='$x(t)$ Analytical', linewidth=2, li
        axes[i, 1].set_title(f'$x(t)$: $t_f = {tf}$')
        axes[i, 1].set_xlabel('$t$')
        axes[i, 1].set_ylabel('$x(t)$')
        axes[i, 1].legend()
        axes[i, 1].grid(True)

        print(f"tf={tf}: max u(t) error = {np.max(np.abs(u_numerical - u_analytical
              f"max x(t) error = {np.max(np.abs(x - x_analytical)):.2e}")

        guess = lambda_0
    plt.tight_layout()
    plt.show()
```

```
solve_and_compare(time_array)
```

```
tf=10: max u(t) error = 1.02e-07, max x(t) error = 4.22e-08
tf=20: max u(t) error = 6.60e-04, max x(t) error = 2.73e-04
tf=30: max u(t) error = 4.36e+02, max x(t) error = 1.80e+02
tf=40: max u(t) error = 6.07e+08, max x(t) error = 2.52e+08
tf=50: max u(t) error = 8.42e+14, max x(t) error = 3.49e+14
```



Interpretation of differences:

What happens to the ability to obtain a solution as tf increases?

Clearly there is a drastic difference between the numerical and analytical solution after $t_f = 20$ - the ability to obtain a solution numerically breaks down around this time. Without using a guess from the prior iteration $t_f = 10$, converges actually breaks down as early as after $t_f = 10$. A solution can be found analytically for all times however. Numerical solutions break down since c_1 is proportional to $e^{-\sqrt{2}t_f}$ which quickly becomes numerically indistingushible to zero from a floating point resolution perspective, causing issues for fsolve. Additionally, the integrator function also gets overwhelmed be the large values from $e^{\sqrt{2}t}$, and there becomes no meaningful way to vary the initial guess lambda(0) to make $x(t_f)$ approach zero due to limits in numerical resolution and a very large variation in x_f from a small perterbation in the initial guess λ_0. The unstable eigenvalue $\sqrt{(}2)$ makes this varation ratio of $\delta\lambda_0$ to $\delta x(t_f)$ only grow larger and larger with t_f making an iterative loop like fsolve be unable to converge properly on the solution.

what do you observe about the structure in the solution as tf increases?

As described in the prior question, c_1 approaches zero and c_2 approaches 1 (sinze c_2 = 1-c_1), thus the c_2 term dominates the initial decay as t increases - numerically this is exacerbated due to resolution limits on representing very small floating points. Integrator error grows proprotionally with t_f as well. Contributions typically made by the unstable c_1 term are either dragged to zero from limitations on floating point resolutions, or are exacerbated massively as a small variation from the true λ_0 has a signifigant contribution at larger t_f values, explaining the fsolve convergance without truly aligned end numerical values - it can't vary λ_0 by a small enough value anymore. It also explains the massive end peaks on each graph - they are a result of a slight variation in λ. This whole discussion mirris ored and applicable to the control u(t) for the same reasons.

Problem 2:

Optimize:

$$J = t_f$$

Subject to:

$$x' = vsin(\theta)$$

$$y' = vcos(\theta)$$

$$v' = gcos(\theta)$$

where g = 10

Boundary conditions:

$$x(0) = x_0 = 0$$

$$y(0) = y_0 = 0$$

$$v(0) = v_0 = 0$$

$$x(t_f) = x_f = 2$$

$$y(t_f) = y_f = 2$$

$$v(t_f) = Free$$

$$t_f = Free$$

Applying to general form:

$$\underline{\mathbf{x}}(t) = \begin{bmatrix} x(t) \\ y(t) \\ v(t) \end{bmatrix}$$

$$u(t) = \theta(t)$$

costate:

$$\underline{\lambda(t)} = (\lambda_x, \lambda_y, \lambda_v)$$

Hamiltonian:

$$H = L + \underline{\lambda}^T \underline{f}(u, x)$$

where L = 0 thus,

$$H = \lambda_x v sin(\theta) + \lambda_y v cos(\theta) + \lambda_v g cos(\theta)$$

Derivation of general optimality conditions is included in appendix as this is already a bloated proof

Applying equations 1.1 - 1.3 from the appendix

$$\lambda_x' = -\frac{\partial H}{\partial x} = 0$$

$$\lambda_y' = -\frac{\partial H}{\partial y} = 0$$

$$\lambda_v' = -\frac{\partial H}{\partial v} = -\lambda_x sin(\theta) - \lambda_y cos(\theta)$$

$$\frac{\partial H}{\partial \theta} = 0 = \lambda_x v cos(\theta) - (\lambda_y v + \lambda_v g) sin(\theta)$$

This tells us λ_x and λ_y is a constant

Revisiting B.C.:

$$x(0) = x_0 = 0$$

$$y(0) = y_0 = 0$$

$$v(0) = v_0 = 0$$

$$x(t_f) = x_f = 2$$

$$y(t_f) = y_f = 2$$

$$v(t_f) = Free$$

$$t_f = Free$$

This also implies:

$$\delta x_i = 0$$

$$\delta x_f = 0$$

$$\delta y_i = 0$$

$$\delta y_f = 0$$

$$\delta v_i = 0$$

$$\delta v_f \neq 0$$

$$\delta t_i = 0$$

$$\delta t_f \neq 0$$

Thus the following optimlatiy conditions applies from (1.4):

$$w_f = \frac{dm}{dt_f} - \mathbf{v}^T \frac{d\mathbf{b}}{dt_f} + H(t_f)$$

where

$$\frac{d\mathbf{b}}{dt_f} = 0$$

so

$$H(t_f) = -1$$

$$H\big|_{t_f} = [\lambda_x v \sin(\theta) + \lambda_y v \cos(\theta) + \lambda_v g \cos(\theta)]\big|_{t_f} = -1$$

$$-\lambda_v(t_f) + \frac{\partial M}{\partial v_f} + \mathbf{v}^T \frac{\partial \mathbf{b}}{\partial v_f} = 0$$

Since $\frac{\partial M}{\partial v_f} = 0$ and $\frac{\partial \mathbf{b}}{\partial v_f} = 0$:

$$-\lambda_v(t_f) = 0$$

So in summary we have the following 4 unkowns:

$$(\lambda_x(0), \lambda_y(0), \lambda_v(0), t_f)$$

And the following 4 equations:

$$H\big|_{t_f} = [\lambda_x v \sin(\theta) + \lambda_y v \cos(\theta) + \lambda_v g \cos(\theta)]\big|_{t_f} + 1 = 0$$

$$-\lambda_v(t_f) = 0$$

$$x(t_f) - x_f = 0$$

$$y(t_f) - y_f = 0$$

With the following constraints:

$$x' = vsin(\theta) \tag{2.1}$$

$$y' = vcos(\theta) \tag{2.2}$$

$$v' = gcos(\theta) \tag{2.3}$$

$$\lambda_x' = -\frac{\partial H}{\partial x} = 0 \tag{2.4}$$

$$\lambda_y' = -\frac{\partial H}{\partial y} = 0 \tag{2.5}$$

$$\lambda_v' = -\frac{\partial H}{\partial v} = -\lambda_x sin(\theta) - \lambda_y cos(\theta) \tag{2.6}$$

$$\frac{\partial H}{\partial \theta} = 0 = \lambda_x v cos(\theta) - (\lambda_y v + \lambda_v g) sin(\theta) \tag{2.7}$$

Code for solving problem 2 with root finding:

In [51]:
```
#imports
from scipy.optimize import root_scalar
import numpy as np
from scipy.integrate import solve_ivp
from scipy.optimize import fsolve
from scipy.optimize import newton

#Global variable decleration:
g = 10.0


def solve_theta_newton(lamx, lamy, lamv, v):

    A = lamx*v
    B = lamy*v + lamv*g

    def F(theta):
        return A*np.cos(theta) - B*np.sin(theta)

    def Fprime(theta):
        return -A*np.sin(theta) - B*np.cos(theta)

    theta0 = 0.0

    return newton(F, theta0, fprime=Fprime, tol=1e-10, maxiter=50)
```

In [52]:
```
def dynamics(t, z):
    x, y, v, lamx, lamy, lamv = z

    # Eq 2.7
    A = lamx * v
    B = lamy * v + lamv * g

    theta = np.arctan(A/B)
    #The below code works and should be used if root finding is perferred but it ha
    #theta = solve_theta_newton(lamx, lamy, lamv, v)
    # End of eq. 2.7

    dxdt = v * np.sin(theta) #Eq. 2.1
    dydt = v * np.cos(theta) #Eq. 2.2
    dvdt = g * np.cos(theta) #Eq. 2.3

    dlamxdt = 0.0 #Constants Eq. 2.4
    dlamydt = 0.0 #Constants Eq. 2.5
    dlamvdt = -lamx*np.sin(theta) - lamy*np.cos(theta) #Eq. 2.6
```

```
    return [dxdt, dydt, dvdt, dlamxdt, dlamydt, dlamvdt]
```

In [53]:
```
def integrator(lamx0, lamy0, lamv0, tf):
    z0 = [0, 0, 0, lamx0, lamy0, lamv0] #initial guess
    sol = solve_ivp(dynamics, [0, tf], z0, rtol=1e-8, atol=1e-8)
    #Given an initial guess will return the integrated
    #values from our dynamics
    return sol.y, sol.t
def error(guess):

    lamx0, lamy0, lamv0, s = guess

    tf = np.exp(s)

    z, t = integrator(lamx0, lamy0, lamv0, tf)

    x_tf   = z[0, -1]
    y_tf   = z[1, -1]
    v_tf   = z[2, -1]
    lamx_tf = z[3, -1]
    lamy_tf = z[4, -1]
    lamv_tf = z[5, -1]

    # Compute theta at final time
    A = lamx_tf * v_tf
    B = lamy_tf * v_tf + lamv_tf * g
    theta_tf = np.arctan(A/B)
    #The below code works and should be used if root finding is perferred but it ha
    #theta_tf = solve_theta_newton(lamx_tf, lamy_tf, lamv_tf, v_tf)

    # Hamiltonian (no +1 inside)
    H_tf = (lamx_tf * v_tf * np.sin(theta_tf)
            + lamy_tf * v_tf * np.cos(theta_tf)
            + lamv_tf * g * np.cos(theta_tf))

    r1 = x_tf - 2
    r2 = y_tf - 2
    r3 = lamv_tf
    r4 = H_tf + 1

    return np.array([r1, r2, r3, r4])
```

In [55]:
```
initial_tf = 0.8165
guess = np.array([0.15, 0.056, -0.1, np.log(initial_tf)])
solution, info, ier, msg = fsolve(
    error,
    guess,
    xtol=1e-8,
    full_output=True
)

lamx0 = solution[0]
lamy0 = solution[1]
lamv0 = solution[2]
```

```
s     = solution[3]
tf = np.exp(s)  # convert back to physical final time
z, t = integrator(lamx0, lamy0, lamv0, tf)
print(f"lambda_x(0) = {lamx0:.10f}")
print(f"lambda_y(0) = {lamy0:.10f}")
print(f"lambda_v(0) = {lamv0:.10f}")
print(f"tf          = {tf:.10f}")
print(f"x(tf)       = {z[0,-1]:.10f}")
print(f"y(tf)       = {z[1,-1]:.10f}")
print(f"lambda_v(tf)= {z[5,-1]:.10f}")
print(f"Converged: {ier==1}, message: {msg}")

guess = solution


# Compute H along the full trajectory
lx, ly, lv, v_arr = z[3], z[4], z[5], z[2]
A = lx * v_arr
B = ly * v_arr + lv * g
theta_traj = np.arctan(A / B)
H_traj = (lx * v_arr * np.sin(theta_traj)
          + ly * v_arr * np.cos(theta_traj)
          + lv * g * np.cos(theta_traj))

print(f"H at tf        = {H_traj[-1]:.10f}")
print(f"Mean H over path        = {np.mean(H_traj):.10f}")
```

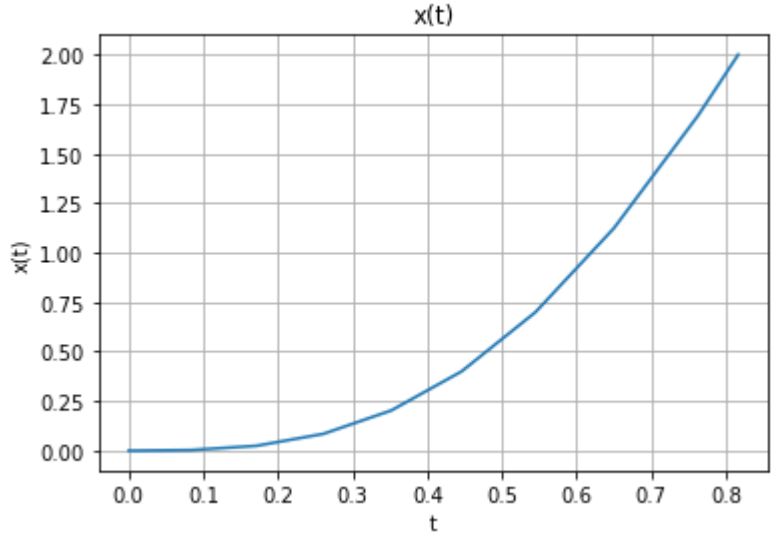
```
lambda_x(0) = -0.1477097414
lambda_y(0) = -0.0564077326
lambda_v(0) = -0.1000000000
tf          = 0.8164698967
x(tf)       = 2.0000000000
y(tf)       = 2.0000000000
lambda_v(tf)= -0.0000000000
Converged: True, message: The solution converged.
H at tf        = -1.0000000000
Mean H over path        = -0.9999999999
```

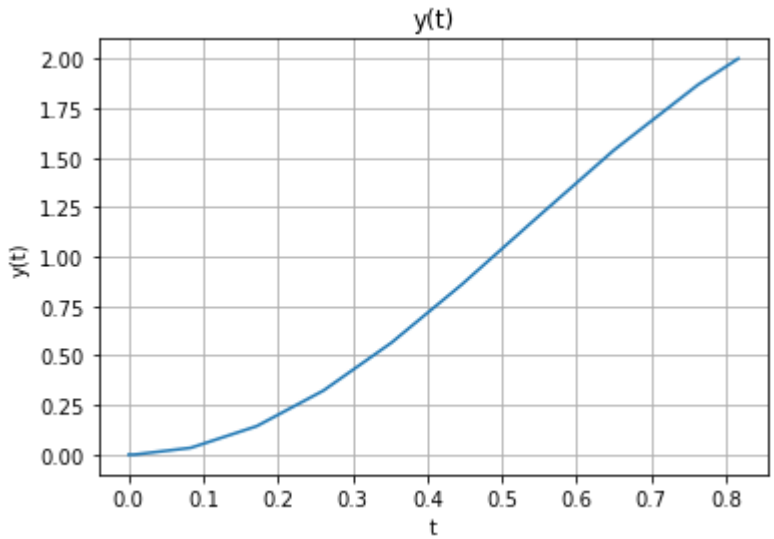
Found trajectories x(t), y(t), v(t), and $\theta(t)$

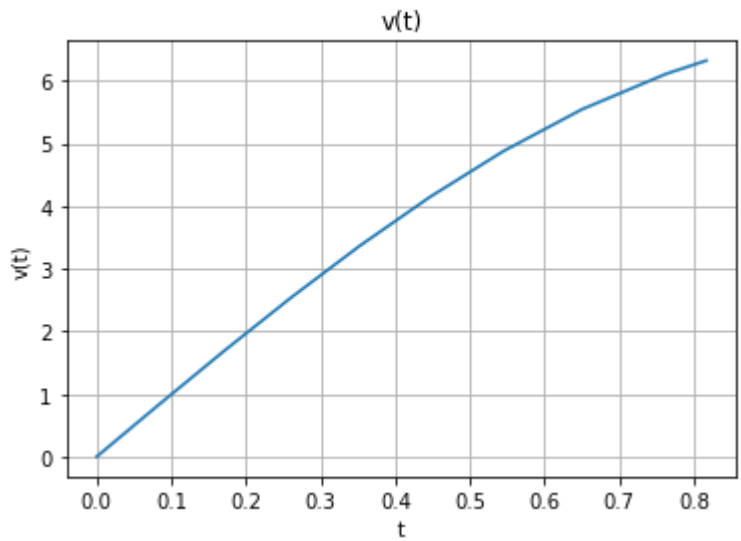
In [58]:

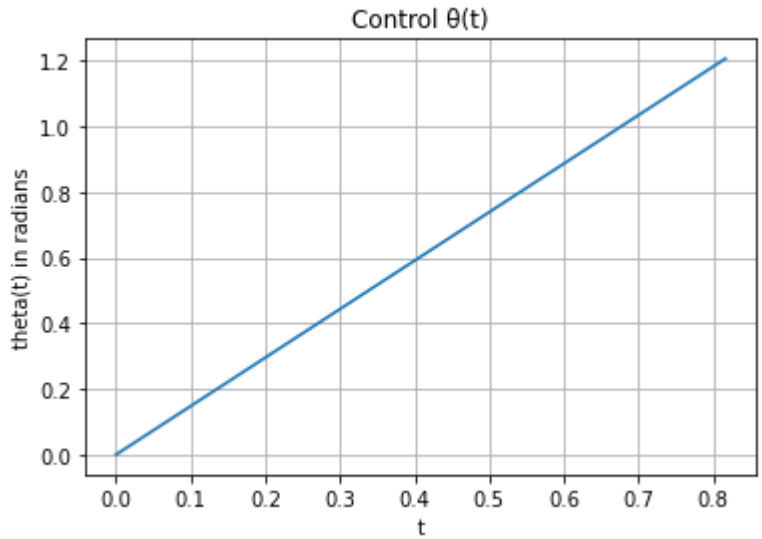
```
import matplotlib.pyplot as plt

x = z[0]
y = z[1]
v = z[2]
lx = z[3]
ly = z[4]
lv = z[5]

A = lx * v
B = ly * v + lv * g
theta = np.arctan(A/B)

# 1) x(t), y(t)
```

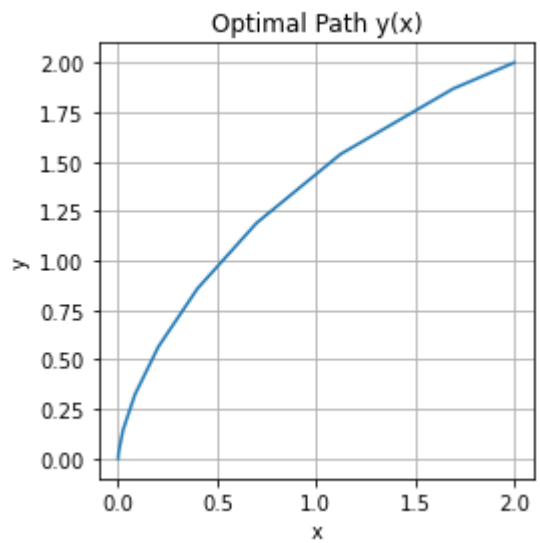
```python
plt.figure()
plt.plot(t, x)
plt.xlabel("t")
plt.ylabel("x(t)")
plt.title("x(t)")
plt.grid()

plt.figure()
plt.plot(t, y)
plt.xlabel("t")
plt.ylabel("y(t)")
plt.title("y(t)")
plt.grid()

# 2) v(t)
plt.figure()
plt.plot(t, v)
plt.xlabel("t")
plt.ylabel("v(t)")
plt.title("v(t)")
plt.grid()

# 3) theta(t)
plt.figure()
plt.plot(t, theta)
plt.xlabel("t")
plt.ylabel("theta(t) in radians")
plt.title("Control θ(t)")
plt.grid()

# 4) spatial curve y(x)
plt.figure()
plt.plot(x, y)
plt.xlabel("x")
plt.ylabel("y")
plt.title("Optimal Path y(x)")
plt.gca().set_aspect('equal', adjustable='box')
plt.grid()

plt.show()
```


x(t)
x(t)
t
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8


y(t)
y(t)
t
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8


v(t)
v(t)
t
6
5
4
3
2
1
0
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8





Appendix: Derivation of General Optimality Conditions

Minimize the **Bolza cost**:

$$J = M\big(\mathbf{x}(t_f),\, t_f\big) + \int_{t_0}^{t_f} L(\mathbf{x},\, u,\, t)\, dt$$

subject to:

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, u, t), \qquad \mathbf{b}\big(\mathbf{x}(t_0), t_0,\, \mathbf{x}(t_f), t_f\big) = \mathbf{0}$$

Append dynamics via costate $\boldsymbol{\lambda}(t)$ and boundary constraints via multipliers $\boldsymbol{\nu}$:

$$J_a = M(\mathbf{x}_f, t_f) - \boldsymbol{\nu}^T \mathbf{b} + \int_{t_0}^{t_f} \big[L + \boldsymbol{\lambda}^T(\mathbf{f} - \dot{\mathbf{x}})\big]\, dt$$

Define the Hamiltonian:

$$\boxed{H(\mathbf{x}, u, \boldsymbol{\lambda}, t) = L(\mathbf{x}, u, t) + \boldsymbol{\lambda}^T \mathbf{f}(\mathbf{x}, u, t)}$$

So that:

$$J_a = M(\mathbf{x}_f, t_f) - \boldsymbol{\nu}^T \mathbf{b} + \int_{t_0}^{t_f} \left[H - \boldsymbol{\lambda}^T \dot{\mathbf{x}} \right] dt$$

Take $\delta J_a = 0$. Integrate $\boldsymbol{\lambda}^T \delta \dot{\mathbf{x}}$ by parts:

$$\int_{t_0}^{t_f} \boldsymbol{\lambda}^T \delta \dot{\mathbf{x}} \, dt = \left[\boldsymbol{\lambda}^T \delta \mathbf{x} \right]_{t_0}^{t_f} - \int_{t_0}^{t_f} \dot{\boldsymbol{\lambda}}^T \delta \mathbf{x} \, dt$$

Collecting terms in $\delta \mathbf{x}$ and δu:

$$\delta J_a = (\text{endpoint terms}) + \int_{t_0}^{t_f} \left[\left(\frac{\partial H}{\partial \mathbf{x}} + \dot{\boldsymbol{\lambda}} \right)^T \delta \mathbf{x} + \frac{\partial H}{\partial u}^T \delta u \right] dt = 0 \tag{1.1}$$

Costate equations (coefficient of $\delta \mathbf{x}$ in integrand):

$$\dot{\boldsymbol{\lambda}} = -\frac{\partial H}{\partial \mathbf{x}} \tag{1.2}$$

Stationarity (coefficient of δu in integrand):

$$\frac{\partial H}{\partial u} = \mathbf{0} \tag{1.3}$$

The endpoint terms arise from three sources: the variation of the Mayer term M, the variation of the boundary constraint term $\boldsymbol{\nu}^T \mathbf{b}$, and the boundary evaluation of the integration by parts. Expanding each:

Variation of $M(\mathbf{x}(t_0), t_0, \mathbf{x}(t_f), t_f)$:

$$\delta M = \frac{\partial M}{\partial \mathbf{x}(t_0)} \delta \mathbf{x}_0 + \frac{\partial M}{\partial t_0} \delta t_0 + \frac{\partial M}{\partial \mathbf{x}(t_f)} \delta \mathbf{x}_f + \frac{\partial M}{\partial t_f} \delta t_f$$

Variation of $\boldsymbol{\nu}^T \mathbf{b}(\mathbf{x}(t_0), t_0, \mathbf{x}(t_f), t_f)$:

$$\delta(\boldsymbol{\nu}^T \mathbf{b}) = \boldsymbol{\nu}^T \frac{\partial \mathbf{b}}{\partial \mathbf{x}(t_0)} \delta \mathbf{x}_0 + \boldsymbol{\nu}^T \frac{\partial \mathbf{b}}{\partial t_0} \delta t_0 + \boldsymbol{\nu}^T \frac{\partial \mathbf{b}}{\partial \mathbf{x}(t_f)} \delta \mathbf{x}_f + \boldsymbol{\nu}^T \frac{\partial \mathbf{b}}{\partial t_f} \delta t_f$$

Boundary evaluation of integration by parts:

$$\left[\boldsymbol{\lambda}^T \delta \mathbf{x} \right]_{t_0}^{t_f} = \boldsymbol{\lambda}^T(t_f) \delta \mathbf{x}_f - \boldsymbol{\lambda}^T(t_0) \delta \mathbf{x}_0$$

Note that when the limits of integration are varied by δt_0 and δt_f, the Hamiltonian evaluated at the endpoints contributes:

$$\left[H \right]_{t_0}^{t_f} \cdot (\text{limit variations}) = H(t_f) \, \delta t_f - H(t_0) \, \delta t_0$$

Combining all endpoint contributions and grouping by variation:

$$\delta J_a\big|_{\text{endpoint}} = \left[\frac{\partial M}{\partial \mathbf{x}_0} - \boldsymbol{\nu}^T \frac{\partial \mathbf{b}}{\partial \mathbf{x}_0} + \boldsymbol{\lambda}^T(t_0)\right] \delta\mathbf{x}_0 + \left[\frac{\partial M}{\partial \mathbf{x}_f} - \boldsymbol{\nu}^T \frac{\partial \mathbf{b}}{\partial \mathbf{x}_f} - \boldsymbol{\lambda}^T(t_f)\right] \delta\mathbf{x}_f$$
$$+ \left[\frac{\partial M}{\partial t_0} - \boldsymbol{\nu}^T \frac{\partial \mathbf{b}}{\partial t_0} - H(t_0)\right] \delta t_0 + \left[\frac{\partial M}{\partial t_f} - \boldsymbol{\nu}^T \frac{\partial \mathbf{b}}{\partial t_f} + H(t_f)\right] \delta t_f = 0$$

Define the four **transversality conditions**:

$$T_0 = \left[\boldsymbol{\lambda}^T(t_0) + \frac{\partial M}{\partial \mathbf{x}(t_0)} - \boldsymbol{\nu}^T \frac{\partial \mathbf{b}}{\partial \mathbf{x}(t_0)}\right]^T$$

$$T_f = \left[-\boldsymbol{\lambda}^T(t_f) + \frac{\partial M}{\partial \mathbf{x}(t_f)} - \boldsymbol{\nu}^T \frac{\partial \mathbf{b}}{\partial \mathbf{x}(t_f)}\right]^T$$

$$W_0 = \frac{\partial M}{\partial t_0} - \boldsymbol{\nu}^T \frac{\partial \mathbf{b}}{\partial t_0} - H(t_0)$$

$$W_f = \frac{\partial M}{\partial t_f} - \boldsymbol{\nu}^T \frac{\partial \mathbf{b}}{\partial t_f} + H(t_f)$$

So that the endpoint condition is written compactly as:

$$T_0^T\, \delta\mathbf{x}_0 + T_f^T\, \delta\mathbf{x}_f + W_0\, \delta t_0 + W_f\, \delta t_f = 0 \tag{1.4}$$

For each term, either the variation is zero (fixed boundary) or the corresponding condition must be zero (free boundary):

- $\delta\mathbf{x}_0 \neq 0 \;\Rightarrow\; T_0 = \mathbf{0}$
- $\delta\mathbf{x}_f \neq 0 \;\Rightarrow\; T_f = \mathbf{0}$
- $\delta t_0 \neq 0 \;\Rightarrow\; W_0 = 0$
- $\delta t_f \neq 0 \;\Rightarrow\; W_f = 0$

In [ ]: